

An Introduction to Ordinary Differential Equations

Chapter 1: Foundations, Geometry, and Separable Systems

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Abstract

This chapter establishes the conceptual and analytical foundations of *ordinary differential equations* (ODEs). We begin by contrasting the philosophy of algebraic equations — which seek unknown *values* — with the dynamic worldview of differential equations — which seek unknown *functions*. We develop precise vocabulary for classification (order, degree, linearity), introduce the geometric language of *slope fields*, derive the method of *separation of variables* from first principles, and apply our tools to model real physical systems. The chapter is paced for a demanding two-hour instructional session and concludes with a comprehensive problem set.

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Module 1: Foundations, Classification, and Verification

0.1 Conceptual Introduction: From Algebra to Dynamics

In your prior mathematical experience, an *equation* has nearly always been an **algebraic** object. The equation

$$x^2 - 5x + 6 = 0$$

poses a question: *What number(s) x satisfy this constraint?* The unknown is a **value** — a static, dimensionless quantity — and our task is to isolate it. This is the world of arithmetic and algebra: the world of *quantities*.

A **differential equation** asks a question of a fundamentally different character. Consider the statement:

$$\frac{dy}{dx} = 2x.$$

The unknown here is not a number. It is a **function**, $y(x)$, whose *rate of change* is specified at every point. We are no longer asking “what value?” but rather “*what dynamic relationship?*”. The answer is itself a rule, a process, a mapping. Solving the above yields not a number but a family of curves $y(x) = x^2 + C$ — infinitely many parabolas, each describing a possible trajectory consistent with the prescribed rate law.

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he philosophical leap from algebraic equations to differential equations is the leap from **statics** to **dynamics**. The natural sciences are written in this language because nature herself does not, in general, hand us finished states; she hands us *laws of change*. Newton did not write down where a planet *is*. He wrote down how its velocity changes (Newton’s Second Law), and the planetary orbit emerged as the *solution* to a differential equation.

Why ODEs Are the Native Language of Science

Almost every quantitative scientific law is, at its heart, a differential equation. A few canonical examples:

- **Physics (Newtonian mechanics):** A falling object under gravity (with air resistance proportional to velocity) obeys

$$m \frac{dv}{dt} = mg - kv.$$

The unknown is the velocity function $v(t)$.

- **Biology (population dynamics):** A population of bacteria with abundant resources grows in proportion to its current size:

$$\frac{dP}{dt} = rP.$$

The unknown is the population function $P(t)$.

- **Finance (continuous compounding):** A bank balance $B(t)$ accruing interest at continuous rate r satisfies

$$\frac{dB}{dt} = rB.$$

Identical mathematical structure — different physical interpretation.

- **Chemistry (radioactive decay):** An unstable isotope decays at a rate proportional to remaining mass:

$$\frac{dN}{dt} = -\lambda N.$$

The unifying observation is striking: *nature repeatedly speaks in the grammar of rate laws*. To do science quantitatively, one must learn to read and write this grammar.

0.2 Core Vocabulary

We now establish the precise terminology that will accompany us throughout the course.

Dependent and Independent Variables

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In a differential equation involving a function $y(x)$, the variable x on which y depends is called the **independent variable**. The function y itself, whose values are determined by x , is called the **dependent variable**.

For example, in $\frac{dy}{dx} = 3x^2$, the independent variable is x and the dependent variable is y . In time-evolution problems such as $\frac{dP}{dt} = rP$, time t is the independent variable and population P is the dependent variable.

Ordinary vs. Partial Differential Equations

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An **ordinary differential equation** (ODE) is a differential equation in which the unknown function depends on a *single* independent variable. A **partial differential equation** (PDE) is one in which the unknown function depends on *two or more* independent variables, and hence involves partial derivatives.

The equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0 \quad (\text{ODE})$$

involves only ordinary derivatives with respect to a single variable. In contrast,

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad (\text{PDE, the heat equation})$$

involves partial derivatives of $u(x, t)$ with respect to two independent variables. This course concerns itself exclusively with ODEs.

Order and Degree

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The **order** of an ODE is the order of the highest derivative that appears in the equation. The **degree** of an ODE is the algebraic power of the highest-order derivative, *after the equation has been rationalized and cleared of fractional and radical expressions involving the derivatives* (and is defined only when such a polynomial form exists).

For example:

- $\frac{dy}{dx} + 2y = x^3$ has order 1 and degree 1.
- $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^5 + y = 0$ has order 2 and degree 3. (*The degree is the power of the highest derivative, not the largest power overall.*)
- $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = y \frac{d^2y}{dx^2}$ must first be rationalized. Squaring yields $1 + (y')^2 = y^2(y'')^2$, which has order 2 and degree 2.

Linearity

This concept will recur throughout the course; we must define it with care.

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An ODE in the dependent variable y and independent variable x is called **linear** if it can be written in the form

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x) y = g(x),$$

where the coefficients $a_0(x), a_1(x), \dots, a_n(x)$ and the forcing term $g(x)$ depend at most on the *independent* variable x . An ODE that cannot be cast in this form is called **nonlinear**.

The defining features of linearity are:

- The dependent variable y and each of its derivatives appear to the **first power only**.
- There are **no products** of y with its derivatives (no $y \cdot y'$, no $(y')^2$, etc.).
- No **transcendental functions** (sine, cosine, exponential, logarithm, etc.) are applied to y or its derivatives.
- Coefficients may be arbitrary functions of x alone.

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linearity is not a cosmetic property — it is the hinge on which an enormous body of theory turns. Linear ODEs obey the *superposition principle*: sums and scalar multiples of solutions are again solutions. Nonlinear ODEs generally do not. The mathematical universe splits, in this single property, into two profoundly different territories.

Examples.

- $y'' - 3y' + 2y = \sin(x)$ is **linear** (coefficients are constants, y and derivatives appear to power 1, $\sin$ is applied only to x).
- $y'' + y y' = 0$ is **nonlinear** (the product $y \cdot y'$ violates linearity).
- $y' + \sin(y) = x$ is **nonlinear** ($\sin(y)$ is a transcendental function of y).
- $x^2 y'' + x y' + y = 0$ is **linear** (variable coefficients $a_i(x)$ are still functions of x alone).

0.3 Solution Spaces: General Solutions, Particular Solutions, and IVPs**The Arbitrary Constant of Integration**

Consider the simple ODE $\frac{dy}{dx} = 2x$. Integrating both sides with respect to x :

$$y(x) = \int 2x \, dx = x^2 + C,$$

where $C \in \mathbb{R}$ is an arbitrary constant. This is not merely a notational nuisance — it carries deep meaning. The single rate law $y' = 2x$ does not uniquely determine $y(x)$; it specifies only the *shape* of the solution (up to vertical translation). To pin down a specific curve, additional information — an *initial condition* — must be supplied.

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The **general solution** of an n th-order ODE is a family of functions containing n arbitrary constants $C_1, C_2, \dots, C_n$, encompassing all possible solutions of the equation. A **particular solution** is any specific member of this family, obtained by assigning numerical values to the constants. An **Initial Value Problem** (IVP) consists of an ODE together with sufficient initial conditions to determine these constants uniquely.

The Initial Value Problem

A first-order IVP has the form

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0.$$

Geometrically, this asks for the unique solution curve that passes through the point (x_0, y_0) in the xy -plane. The existence and uniqueness of such a solution (under mild regularity conditions on f) is the content of the **Picard–Lindelöf theorem**, which we shall revisit later in the course.

For an n th-order ODE, an IVP requires n conditions, typically of the form

$$y(x_0) = y_0, \quad y'(x_0) = y'_0, \quad \dots, \quad y^{(n-1)}(x_0) = y_0^{(n-1)}.$$

A Concrete Illustration

Suppose $y'(x) = 2x$ with the initial condition $y(1) = 5$. The general solution is $y = x^2 + C$. Imposing the initial condition:

$$5 = (1)^2 + C \implies C = 4.$$

The particular solution is therefore $y(x) = x^2 + 4$. The infinite family has collapsed to a unique curve.

Module 1 Exercises. Work each problem carefully on a separate sheet; full justification is expected.

(A) Classification (state order, degree, and linearity for each).

1.1 $\left(\frac{d^3y}{dx^3}\right)^2 + 5\left(\frac{dy}{dx}\right)^4 + 7y = \cos(x).$

1.2 $\frac{d^2y}{dx^2} + e^x \frac{dy}{dx} + y \sin(x) = x^3.$

1.3 $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{d^2y}{dx^2}. \quad (\text{Hint: rationalize first.})$

(B) Verification.

1.4 Verify by direct substitution that $y(x) = e^{-2x} \sin(3x)$ is a solution of

$$y'' + 4y' + 13y = 0.$$

1.5 Verify that the implicit relation $x^2 + y^2 = C$ defines a family of solutions to the ODE

$$\frac{dy}{dx} = -\frac{x}{y}.$$

Module 2: Graphical Intuition — Slope Fields

0.4 Geometry Before Algebra

Before we develop analytical techniques to *solve* ODEs, we should ask: *what does the solution look like, geometrically?* For a first-order ODE

$$\frac{dy}{dx} = f(x, y),$$

the right-hand side $f(x, y)$ assigns, to every point (x, y) in the plane, a **slope**. The differential equation is, in this sense, a *rule for slopes*: a recipe that tells the solution curve which direction to head at every point it might possibly visit.

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slope field (or **direction field**) for the first-order ODE $\frac{dy}{dx} = f(x, y)$ is a diagram in the xy -plane in which, at a representative array of points (x, y) , a short line segment is drawn with slope $f(x, y)$. The collection of such segments visually approximates the family of solution curves: any solution $y(x)$ must, at each point on its graph, be tangent to the corresponding segment.

The Power of Geometry

This visualization is powerful for several reasons:

1. **It works even when analytical solution is impossible.** Many ODEs (e.g. $y' = \sin(xy)$) have no closed-form solution, yet their qualitative behavior is fully captured by the slope field.
2. **It reveals equilibrium solutions.** A constant function $y(x) = c$ is a solution precisely when $f(x, c) = 0$ for all x . Such equilibria appear as horizontal flow lines in the slope field.
3. **It exposes long-term behavior.** Stability, divergence, and asymptotic behavior are often visible *at a glance*.

Reading a Slope Field

A few patterns recur often enough to deserve memorization:

- If $f(x, y)$ depends only on x (e.g. $y' = x$), the slope is constant along every vertical line. Solution curves are vertical translates of one another.
- If $f(x, y)$ depends only on y (e.g. $y' = y$ or $y' = y(1 - y)$), the slope is constant along every horizontal line. The ODE is then said to be **autonomous**, and equilibrium solutions are horizontal lines.

- Wherever $f(x, y) = 0$, the slope segment is horizontal — a candidate location for an equilibrium or a local extremum of a passing solution.
- Wherever $|f(x, y)| \rightarrow \infty$, segments become nearly vertical: solution curves change rapidly there.

An Illustrative Example

Consider $\frac{dy}{dx} = y$. The slope at any point equals the y -coordinate of that point. Therefore:

- On the line $y = 0$, all slopes are 0 — horizontal segments. $y = 0$ is an equilibrium solution.
- For $y > 0$, slopes are positive and grow with y : solutions accelerate upward.
- For $y < 0$, slopes are negative: solutions accelerate downward.

Without solving anything, we deduce that $y = 0$ is an unstable equilibrium and that all non-equilibrium solutions diverge exponentially — which is exactly correct, since the general solution is $y = Ce^x$.

Module 2 Exercises.

2.1 Consider the slope field generated by the ODE

$$\frac{dy}{dx} = y - 2.$$

(a) Identify all equilibrium solutions. (b) Describe the qualitative behavior of solutions for $y > 2$ and for $y < 2$. (c) Is the equilibrium stable or unstable? Justify geometrically.

2.2 Sketch (in rough qualitative form) the slope field of

$$\frac{dy}{dx} = -\frac{x}{y}, \quad y \neq 0,$$

on the region $-3 \leq x \leq 3$, $-3 \leq y \leq 3$. Identify the geometric family of solution curves implied by the pattern. (*Hint: compare the slope at a point (x, y) to the slope of the line from the origin to that point. What family of curves has tangents perpendicular to radial lines?*)

Module 3: Analytical Methods — Separable Equations

0.5 First-Principles Logic of Separation of Variables

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first-order ODE is called **separable** if it can be written in the form

$$\frac{dy}{dx} = g(x) h(y),$$

that is, if the right-hand side factors as a product of a function of x alone and a function of y alone.

Why Separation Is Logically Justified

The colloquial procedure — “move all the y ’s to one side, all the x ’s to the other, and integrate” — can appear cavalier, because the symbol $\frac{dy}{dx}$ is technically a *single object* (a limit of difference quotients), not a literal fraction. The legitimacy of the procedure rests on the **Chain Rule**, as follows.

Starting from

$$\frac{dy}{dx} = g(x) h(y),$$

assume $h(y) \neq 0$ in the region of interest. Divide both sides by $h(y)$:

$$\frac{1}{h(y)} \frac{dy}{dx} = g(x). \quad (*)$$

Now define an antiderivative $H(y) = \int \frac{1}{h(y)} dy$. By the Chain Rule,

$$\frac{d}{dx}[H(y(x))] = H'(y) \frac{dy}{dx} = \frac{1}{h(y)} \frac{dy}{dx}.$$

So the left-hand side of $(*)$ is exactly $\frac{d}{dx}[H(y(x))]$. Hence $(*)$ becomes

$$\frac{d}{dx}[H(y(x))] = g(x).$$

Integrating both sides with respect to x :

$$H(y(x)) = \int g(x) dx + C.$$

Returning to Leibniz notation, this is the same outcome as if we had “multiplied through by dx and integrated termwise”:

$$\int \frac{dy}{h(y)} = \int g(x) dx + C.$$

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he manipulation “ dy and dx as separable objects” is therefore not a sloppy shortcut — it is a faithful shorthand for an application of the Chain Rule. *Provided* $h(y) \neq 0$, the procedure is rigorously valid. We must, however, separately check the case $h(y) = 0$: such roots typically correspond to additional **equilibrium solutions** that the algebraic procedure may have suppressed.

0.6 Step-by-Step Worked Example

Solve the initial value problem

$$\frac{dy}{dx} = \frac{x}{y}, \quad y(0) = 2.$$

Step 1: Verify separability. The right-hand side factors as $\frac{x}{y} = x \cdot \frac{1}{y}$, with $g(x) = x$ and $h(y) = \frac{1}{y}$. The equation is separable.

Step 2: Separate the differentials. Multiply both sides by $y \, dx$:

$$y \, dy = x \, dx.$$

The variables are now isolated on opposite sides.

Step 3: Integrate both sides.

$$\int y \, dy = \int x \, dx \quad \implies \quad \frac{y^2}{2} = \frac{x^2}{2} + C_1.$$

Multiplying through by 2 and renaming the constant $C = 2C_1$:

$$y^2 = x^2 + C.$$

This is the **general solution** in implicit form.

Step 4: Apply the initial condition. Substitute $x = 0$, $y = 2$:

$$(2)^2 = (0)^2 + C \quad \implies \quad C = 4.$$

The particular solution (still implicit) is

$$y^2 = x^2 + 4.$$

Step 5: Solve explicitly for y (if desired). Since $y(0) = 2 > 0$, we select the positive

branch:

$$y(x) = \sqrt{x^2 + 4}.$$

Step 6: Verify. We compute $y' = \frac{x}{\sqrt{x^2 + 4}} = \frac{x}{y}$, confirming the ODE. Also, $y(0) = \sqrt{4} = 2$, confirming the initial condition. ■

0.7 Guided Practice Problems

Module 3 Practice. Solve each ODE by separation of variables. Show all steps, and where an initial condition is given, find the particular solution.

3.1 (Polynomial separation). $\frac{dy}{dx} = 3x^2 y.$

3.2 (Exponential separation). $\frac{dy}{dx} = e^{x-y}. \quad (\text{Hint: } e^{x-y} = e^x e^{-y}.)$

3.3 (Rational/trigonometric mixed). $\frac{dy}{dx} = \frac{\cos(x)}{1+y^2}, \quad y(0) = 0.$

3.4 (Secant-polynomial mix, IVP). $(1+x^2) \frac{dy}{dx} = x y, \quad y(0) = 3.$

Module 4: Real-World Application Modeling

0.8 Translation Mechanics: From Words to Rate Laws

The hardest step in applying ODEs is often *not* the solving — it is the **formulation**: rendering a verbal physical law into a precise rate equation. The following mental dictionary will serve you reliably:

- “The rate of change of Q with respect to time” $\longleftrightarrow \frac{dQ}{dt}$.
- “... is proportional to Q ” $\longleftrightarrow = kQ \quad (k > 0)$.
- “... is proportional to Q , but decreasing” $\longleftrightarrow = -kQ \quad (k > 0)$.
- “... is proportional to the difference between Q and an ambient value A ” $\longleftrightarrow = k(Q - A)$ or $-k(Q - A)$ depending on direction.
- “Initially $Q = Q_0$ ” $\longleftrightarrow Q(0) = Q_0$.

Three steps suffice in nearly every modeling problem:

1. **Identify the dependent variable** (what is changing?) and the **independent variable** (with respect to what?).
2. **Translate the physical law** into a differential equation using the dictionary above.
3. **Solve** (typically by separation), then **apply initial/auxiliary data** to determine constants, then **interpret** the result in physical terms.

0.9 Exemplar: Newton’s Law of Cooling

Problem. A cup of coffee is removed from a brewing pot at 95°C and placed in a room held at 20°C . After 10 minutes, the coffee has cooled to 80°C . According to Newton’s Law of Cooling, the rate at which the coffee’s temperature changes is proportional to the difference between its temperature and that of its surroundings.

- (a) Formulate the ODE governing the coffee’s temperature $T(t)$.
- (b) Solve to find $T(t)$ in closed form.
- (c) At what time t^* does the coffee reach 50°C ?

Step 1: Formulate the model. Let $T(t)$ denote the temperature ($^\circ\text{C}$) at time t (minutes). Newton’s Law of Cooling translates directly:

$$\frac{dT}{dt} = -k(T - 20), \quad k > 0,$$

with initial condition $T(0) = 95$ and auxiliary datum $T(10) = 80$. The sign is negative because the coffee is hotter than the room and therefore cooling.

Step 2: Separate and integrate.

$$\begin{aligned}\frac{dT}{T-20} &= -k dt &\implies \int \frac{dT}{T-20} &= \int (-k) dt \\ &&\implies \ln|T-20| &= -kt + C_1.\end{aligned}$$

Exponentiating and absorbing signs into a new constant A :

$$T(t) - 20 = A e^{-kt}, \quad A \in \mathbb{R}.$$

Step 3: Apply $T(0) = 95$ to find A .

$$95 - 20 = A e^0 \implies A = 75.$$

So $T(t) = 20 + 75 e^{-kt}$.

Step 4: Apply $T(10) = 80$ to find k .

$$80 = 20 + 75 e^{-10k} \implies e^{-10k} = \frac{60}{75} = \frac{4}{5}.$$

Taking natural logarithms:

$$-10k = \ln\left(\frac{4}{5}\right) \implies k = \frac{1}{10} \ln\left(\frac{5}{4}\right) \approx 0.02231 \text{ min}^{-1}.$$

Step 5: Solve for t^* when $T = 50$.

$$50 = 20 + 75 e^{-kt^*} \implies e^{-kt^*} = \frac{30}{75} = \frac{2}{5}.$$

Therefore

$$t^* = -\frac{1}{k} \ln\left(\frac{2}{5}\right) = \frac{\ln(5/2)}{\ln(5/4)} \cdot 10 \approx 41.07 \text{ minutes}.$$

Interpretation. The coffee reaches 50°C roughly 41 minutes after being placed in the room. Note that as $t \rightarrow \infty$, $T(t) \rightarrow 20$: the coffee approaches but never quite reaches room temperature, exactly as we would expect physically. ■

0.10 Practice Problem

4.1 (Bacterial growth, model from scratch).

A bacterial colony grows at a rate proportional to its current population. At time $t = 0$ (in hours) the colony numbers 500 cells. Three hours later, the colony has grown to 1500 cells.

(a) Write down the IVP governing the population $P(t)$.

- (b) Solve for $P(t)$ explicitly.
- (c) How long, from $t = 0$, does it take for the colony to reach 10,000 cells?
- (d) What is the **doubling time** of this colony?

Problem Set 1: Foundations & Separable Systems

This homework assignment consolidates the techniques and concepts of Modules 1–4. Show all work; full credit requires complete justification, not merely correct final answers. Where an initial condition is supplied, provide the particular solution. Where one is not, the general solution (with explicit arbitrary constants) suffices.

Part A: Classification (5 problems)

For each ODE below, state the (i) order, (ii) degree, and (iii) whether the equation is linear or nonlinear. Justify your linearity determination by citing the specific violating term where applicable.

PS1.1 $\frac{d^3y}{dx^3} + 2x \frac{d^2y}{dx^2} + y = \sin(x).$

PS1.2 $\left(\frac{dy}{dx}\right)^2 + xy = e^x.$

PS1.3 $\frac{d^4y}{dx^4} - y \frac{dy}{dx} = 0.$

PS1.4 $\sqrt{\frac{dy}{dx}} + y = x. \quad (\text{Rationalize before assigning a degree.})$

PS1.5 $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (x^2 - 1)y = 0. \quad (\text{This is Bessel's equation of order 1.})$

Part B: Solution Verification (3 problems)

Verify by substitution that each proposed function is a solution of the given ODE. For PS1.8, also confirm the initial conditions.

PS1.6 Verify that $y(x) = C e^{2x}$ is a solution of $y' - 2y = 0$ for any constant $C \in \mathbb{R}$.

PS1.7 Verify that $y(x) = x^2 + \frac{C}{x}$ is a solution of the linear ODE

$$x y' + y = 3x^2.$$

PS1.8 (IVP). Verify that $y(x) = 2e^{3x} - e^{-x}$ is the solution of the initial value problem

$$y'' - 2y' - 3y = 0, \quad y(0) = 1, \quad y'(0) = 7.$$

Part C: Separable ODEs (5 problems)

Solve each equation by separation of variables. State the general solution explicitly when possible, or implicitly when explicit isolation is impractical.

PS1.9 $\frac{dy}{dx} = xy.$

PS1.10 (IVP). $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}, \quad y(0) = 1.$

PS1.11 $\frac{dy}{dx} = e^{x-y}.$

PS1.12 $\frac{dy}{dx} = \frac{x^2}{y(1+x^3)}.$

PS1.13 (IVP). $(1+x^2)\frac{dy}{dx} = 4xy, \quad y(0) = 5.$

Part D: Real-World Modeling (2 problems)

For each problem: (i) define your variables explicitly, (ii) formulate the ODE and initial conditions, (iii) solve, and (iv) interpret your answer in physical terms.

PS1.14 (Radioactive decay). Carbon-14, an isotope used in radiocarbon dating, decays at a rate proportional to the remaining mass. Its half-life is 5,730 years.

- (a) Write down the IVP describing $N(t)$, the mass of C^{14} remaining at time t years, starting from an initial mass N_0 .
- (b) Solve for $N(t)$ in terms of N_0 .
- (c) Determine the decay constant λ explicitly, using the half-life condition.
- (d) A wooden artifact is found to contain 30% of the C^{14} that a living tree of the same species would contain. Estimate the age of the artifact.

PS1.15 (Mixing problem). A tank initially holds 200 litres of pure water. Brine containing 0.05 kg of salt per litre flows into the tank at a constant rate of 4 L/min, and the well-stirred mixture flows out at the same rate 4 L/min.

- (a) Let $S(t)$ denote the kilograms of salt in the tank at time t (minutes). Explain in one sentence why the volume of liquid in the tank remains constant at 200 L.
- (b) Argue that the rate of change of salt is

$$\frac{dS}{dt} = (\text{rate in}) - (\text{rate out}) = 0.05 \cdot 4 - \frac{S(t)}{200} \cdot 4.$$

Simplify this to obtain a separable ODE in S .

- (c) Solve the IVP with $S(0) = 0$.
- (d) Determine the long-time limiting value $\lim_{t \rightarrow \infty} S(t)$ and interpret it physically.

End of Problem Set 1. Total: 15 problems across four parts. Recommended preparation time: 3–4 hours.